



CIVIL MAINT. Department
SIMHADRI



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Subject: Approval of LMI on Guidelines for INSPECTION AND
MAINTENANCE OF STEEL AND CONCRETE STRUCTURES (LMI no.
LMI/OGN/OPS/CIV/007)

Subject: Approval of LMI on Guidelines for INSPECTION AND MAINTENANCE OF STEEL AND
CONCRETE STRUCTURES (LMI no. LMI/OGN/OPS/CIV/007)

LMI references:

1. OGN/OPS/CIV/007 rev.2, May 2023: Copy enclosed

Based on the above mentioned OGN (revised in May 2023), the subject LMI has been reviewed and revised accordingly (Rev 8).

Submitted for approval.

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LMI for Inspection and Maintenance of Steel and Concrete Structures has been reviewed in line with Revision No- 2 of COS-ISO-00-OGN/OPS/CIV/007 and submitted for approval pl.

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Local Management Instruction

*Inspection and Maintenance of Steel
and Concrete Structures
(Non-Corrosive Atmosphere)*

LMI/OGN/OPS/CIV/007



NTPC Simhadri

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	INSPECTION AND MAINTENANCE OF STEEL & CONCRETE STRUCTURES
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/CIV/007/</u>
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1.0 SUMMARY

1.1. Superseded Document	Operation Guidance Note – Inspection and Maintenance of Steel and Concrete Structures (Non-Corrosive Atmosphere): OGN/OPS/CIV/007 Rev. No. 1 dated 03/11/2020	
1.2. Scope	The scope of this LMI is to detail the methods that are to be adopted for the inspection and maintenance of all steel & concrete structures inside and outside plant area (in non-corrosive atmospheres). Inspection of chimney and structures in the vicinity of corrosive chemicals are not covered in this LMI.	
1.3. Review	Review Responsibility	O&M Civil
	Review Frequency	2 yearly basis, or as and when required, whichever is lower
	Review Approval	BUH Simhadri

2.0 INTRODUCTION

At Simhadri the structures are engineered and erected for the most critical combinations of dead loads, imposed loads, equipment loads, crane loads, piping loads (static & dynamic), wind loads, seismic loads and temperature loads. In addition, erection loads have also been considered. Dead loads include the weight of structures complete with finishes, fixtures and partitions. Imposed loads include live loads, minor equipment loads like MCC and control panel, erection, operation and maintenance loads. Loads of equipment viz. Deaerator, HP/LP heater, feed water tank, BFP, motors, conveyor loads, coal loads, tripper, chutes, monorails, cable loads, tanks, ducts, mills, coal feeders etc. are taken as given by manufacturer are taken over and above the imposed loads. For crane loads an impact factor and lateral surge loads are also considered. Site specific seismic information has been adopted for calculating seismic forces. All structures are designed for wind forces taking site specific wind parameters (on peak wind speed method; 3sec gust) into consideration. For temperature loading the total temperature variation has been considered as 2/3 of the average minimum annual variation in temperature. Switch yard structures have been designed apart from the worst combination of above-mentioned loads, loads due to deviation of conductor, loads due to unbalanced tension in conductor, torsional load due to unbalanced vertical and horizontal forces, short circuit forces including snap in case of bundled conductors. Boiler support structures are so configured that the temperature of steel does not exceed 60C.

All the buildings at Simhadri have framed super structures. While main plant building, bunker building, transfer tower, conveyor galleries and trestles, boiler, ESP, Duct work have structural steel frame work, service building and switch yard control room have RCC frame work.

All equipment/structures have been supported on pile foundations or on open foundations (open foundations only for lightly loaded structures where loading intensity is less than 5T/sq.m). The concrete sub structure has been designed based on aggressiveness of soil, chemical analysis of soil & sub-soil water. The protection of super structures have been designed considering the exposure condition and degree of atmospheric corrosion. The painting schedule is epoxy based and anticipated maintenance free life of coating is 05 years.

3.0 RECORDS OF REVISION

- 3.1 Areas that are to be inspected are listed with responsibility groups (Point 5).
- 3.2 Responsibility of inspection of CHP Structures changed from Head CHP Mechanical Maintenance to HOD FM (Point 5).
- 3.3 MTP department defined as Responsibility center for Coordination of inspection and owner of subject OGN (point 5).
- 3.4 Amendment and Clarification in the Frequency of inspection due to recent structure failures in the new stations as well as in the old stations. Earlier it was at the end of 10th, 20th & 25th year & every 05 years thereafter. The revision is proposed as herewith (Point 6)
- 3.5 Internal Inspection frequency has been revised to one year uniformly for all the areas of Plant after COD.
- 3.6 Every 5th year external agency inspection of CHP, Cooling Towers. Any additional area for external inspection to be identified during the yearly internal inspection.
- 3.7 Annexure I "Check List" amended.
- 3.8 Annexure II "Area Wise Inspection Record & Action Plan" for review in ORT/ROPR added.
- 3.9 Annexure III "A Reference report" added.

4.0 AREAS FOR INSPECTION

4.1. Steel Structure

4.1.1 The following shall be inspected/checked in steel structure

- a) Damage to joints due to loosening of permanent bolts or cracks in welds.
- b) Reduction of cross-sectional area due to rusting (corrosion) or abrasion.
- c) Excessive deflection/buckling of member or structure.
- d) Damage (breakdown) to protective coating.
- e) Free movement of expansion/sliding joints.
- f) Part cutting of a section, or a Crack noticed in any member.

- g) Removed/missing member.
 - h) Transmission of vibration to the structures from machines or other moving parts, which may cause fatigue of structural members.
 - i) Any accumulation of foreign materials like dust, ash, scrap etc. on structure (especially on bottom chord of conveyor galleries) should not be allowed.
 - j) Any accumulation of ash on building roof and growth of vegetation must be watched and cleaned especially before monsoon.
 - k) In BOP areas particularly CHP areas in TP and conveyors, where 24Hrs manpower is deployed, urination caused corrosion is prevalent which has caused certain serious structure failures. These areas are to be found and corrective actions taken.
- 4.1.2 Joints in steel structures are vulnerable to developing faults, particularly loosening of bolts. Spring washers, tab washers or locknuts may prevent the problem from occurring. Welded joints may crack due to cyclic loading or poor initial preparation and welding methods.
- 4.1.3 Due to corrosion and rust formation, the effective cross - sectional area of a member may get significantly reduced. The reduction of the cross-sectional area also takes place due to mechanical abrasion with some other moving body or substance.
- 4.1.4 The rate of corrosion depends largely on the type of atmosphere to which the structure is exposed. Industrial pollutants e.g. Sulphur Dioxide, or Sodium Chloride in Marine environments influence the rate at which steel corrodes. Additionally, exposure to moisture, e.g. external structures, can rapidly lead to corrosion.
- 4.1.5 Sometimes the provision of a sacrificial thickness of metal is added to cater for a corrosion margin for the areas where protective coating is either not provided or regular maintenance is not feasible. The inside face of C.W. liner, chimney liner, bunker etc. are few example where allowance of a sacrificial corrosion thickness is made.
- 4.1.6 Similarly, the movement of coal in a coal bunker generates abrasion and erodes the sectional thickness. This is partially overcome by providing an abrasion resistant liner or by providing a sacrificial thickness or metal.
- 4.1.7 Excessive deflection/buckling of a member may take place due to overloading/ stressing/ collision with some heavy earth moving equipment or some other reasons, e.g. overloading may be due to excessive accumulation of dust on the roof/floor, particularly for the coal handling plant area, storage of materials/components for maintenance purpose on areas not specified, use of areas for laydown or unloading purposes not designated for his purpose. Routine inspection and cleaning is essential to avoid excessive loading.
- 4.1.8 Steel structures are provided with a protective coating to protect against corrosion or for a better appearance. Metallic coatings or organic paints are used for this purpose.
- 4.1.9 Galvanization is the most common form of metallic coating. Organic paint systems consist of a primer and finishing coat. For non-corrosive (mild) atmosphere condition, red-oxide zinc chromate is generally used as a primer and synthetic enamel paints is

provided as a finishing coat. For corrosive atmospheric conditions, better quality chemical resistant paints, e.g., chlorinated rubber, epoxy, resin, polyurethane, zinc silicate etc., are used. Regular inspection and maintenance of the painting system is essential for prolonging the life of the coating besides providing adequate protection of structure steel.

- 4.1.10 The expansion/sliding joints are provided to cater for thermal expansion and to avoid the transfer of horizontal forces (due to wind loading, seismic activity, etc.). Neoprene-rubber bearings, PTFE sliding bearings, graphite lubricated sliding bearings and metallic bearings are examples of commonly used bearings. Structural members are provided with slotted holes to allow for any movement. Joints may seize due to the accumulation of dust or corrosion. Regular inspection and maintenance is required for these joints.
- 4.1.11 Some structural members like secondary beams, plan/vertical bracings, member of a truss, etc., may sometimes be cut and removed at site to facilitate routing or maintenance of some facilities without restoring those members in position. These also include removable type of member. This problem leads to structural instability, particularly under horizontal loads. Occasionally, instead of removal of the complete member, portion of a member (like flanges of beam/column, bracing, lacing/batten, member of truss) is locally cut and removed at site to generate additional passage without strengthening the members. This reduces the sectional capacity of the members.
- 4.1.12 Any structure to be cut only after written permission from EIC and should be restored by qualified structural welder only.
- 4.1.13 If structural steel work has to be cut or removed to allow lifting or movement of heavy plant items, then an assessment of the effect on the structure must be made by a qualified Structural Engineer. Temporary steel work may have to be used to strengthen the structure during this type of work. If any heavy components are to be lifted from a steel structure (other than recognized lifting beams) then an assessment must be made of the strength of the structure prior to the lifting work and the lift planned to ensure that no component is subject to stress beyond its design limits.
- 4.1.14 Vibration is occasionally transmitted to a structure from rotating machines or any other vibrating part. Vibration of the coal handling plant gallery, crane girder, floor of crusher house, etc., are very common. A higher amplitude of vibration is not only uncomfortable for the occupants, but it affects the durability of the structures, particularly the fatigue of welded joints. These higher amplitude vibrations may also sometimes get transmitted to the adjoining building structures where proper isolation gap is not maintained.

4.2. Concrete Structures

- 4.2.1 The following will be inspected and checked in concrete structure.
 - a) Condition of Roof water proofing and signs of water accumulation.
 - b) Spalling of concrete or exposure of reinforcement (*saria*).
 - c) Protruding of reinforcement bars from Concrete.

- d) Cracks in concrete or associated Brick walls.
 - e) Seepage/leakage through concrete
 - f) Excessive deflection
 - g) Settlement/differential settlement of the structures.
 - h) Corrosion of reinforcement steel
 - i) Any damaged member liker column/beam/slab
 - j) Abrasion of concrete surfaces
 - k) Honeycombing in concrete structure
 - l) Painting condition of concrete of water retaining structures like CTs, DM/PT plants.
- 4.2.2 Disintegration of concrete: mostly due to attack by sulphate, nitrates and other salts in solution, on the hardened cement paste. Disintegration of concrete is mostly associated with underground concrete structures in the presence of sulphate in the soil/ground water. In most of the cases, this type of attack cannot be inspected as the structure is covered with earth or water.
- 4.2.3 Sulphur compounds (Sulphur dioxide gas, etc.) may react and form sulphuric acid leading to disintegration of concrete, if the attack continues. Sulphur attack can be recognized by white or yellowish coloration of concrete.
- 4.2.4 **Cracks in hardened concrete:** May be of the structural or non-structural types. Structural cracks may be due to accidental overload, creep and under-design of the structure. Non-structural cracks may be of a physical, chemical or thermal in nature. Shrinkable/expansive aggregate, drying shrinkage and crazing are responsible for physical type cracks. Corrosion of reinforcement, alkali-aggregate reactions and cement carbonation are covered under chemical nature of cracks. Freeze/Thaw cycles, external seasonal temperature variations and early thermal contraction (due to external restraint and internal temperature gradients) are the causes of thermal cracks.
- 4.2.5 The cause of the crack shall be ascertained from the pattern and nature of the cracks. Wherever required, testing of the concrete shall be carried out, e.g., ultrasonic testing, rebound hammer test, testing of core for strength. pH value, chloride content, etc.
- 4.2.6 **Seepage/leakage of water:** Particularly in retaining structures and basement type structures (e.g. track hopper, coal handling plant tunnel) shall be checked to ascertain water loss or maintaining dry working conditions. This leakage is generally through the construction/expansion joint. Location and cause of any leakage shall be identified before preparation of rectification scheme.
- 4.2.7 **Excessive deflection:** The deflection of some important structural members e.g. cantilever beams/slabs, long span beams, waffle slabs, folded plates, cylindrical shell roofs, etc., shall be checked. Excessive deflection may be due to overloading, inadequate design or improper detailing.

- 4.2.8 **Settlement/differential settlement:** The problem of settlement/differential settlement of foundations is associated with soil type of foundation design. Differential settlement leads to additional loads/forces in the structure, which may make the structure unsafe. This problem occurs mostly in open type shallow foundations resting on compressible strata.
- 4.2.9 **Corrosion of Reinforcement :** Reinforcement steel embedded in concrete does not normally corrode because in the inherently alkaline environment having pH value well above 12 for good concrete, a protective passive layer is formed on its surface. However, if the depth of cover is insufficient or the concrete is permeable, then the concrete may carbonate (due to the action of carbon dioxide present in the air) and the protective layer may be at risk. The passive layer would also break down in the presence of excessive amount of chloride ions. The chlorides can originate from sodium chloride (common salt) in marine locations or chloride present within the concrete due to water and added chemicals used in the concrete. Corrosion of reinforcement can take place due to oxidation under damp conditions. Oxygen and moisture can enter the concrete through its capillary pores and act on the reinforcement surface.
- 4.2.10 One of the symptoms of reinforcement corrosion is the cracking of concrete in line with the steel. As the corrosion progressive, the concrete begins to spall and exposes the corroded steel. In less advanced cases, rust stains can be seen at the surface. Iron oxide or rust has a volume greater than twice the metallic iron from which it is formed. Corrosion of reinforcement takes place without much increase in the volume when magnetite (Fe_3O_4) is formed due to further oxidation of localized area.
- 4.2.11 **Abrasion of the concrete surfaces:** Generally, occur in the Track hopper, silos, etc., due to flow of coal. Lining with Guniting is generally provided to protect the base concrete. Excessive wear of the concrete would expose the reinforcement and reduce the sectional thickness.
- 4.2.12 **Expansion joint damages:** Expansion joints are provided to cater for expansion and contraction of structures. Sometimes the gap gets filled with some non-compressible material. Similarly, in case of isolation gaps provided around vibrating foundations may also get filled. Thus, the proper functioning of the joints/ gaps gets restricted.

5.0 RESPONSIBILITY

The primary responsibility of inspection of steel and concrete structure will be of the maintenance group looking after the concerned area. The overall responsibility will be:

- Head of Fuel Management for CHP
- Head of Maintenance for the rest of the plant.

The MTP Dept will be responsible for this OGN implementation and the focal point for all the structural assessment works. If any corrective action are taken, individual sections will generate reports on these and submit to MTP for record keeping. MTP be custodian of these inspection/repair reports and upload these

on its intranet webpage.

List of areas to be inspected and maintenance group responsible are as follows:

Areas	Responsible Group
Coal Handling Plant (including unloading facilities, TPs, Conveyors, Crusher Houses, pump house etc)	CHP O&M Group
TG Hall & associated building including control room.	TG Maintenance Group
Boiler and ESP	Boiler Maint Group
Ash Silos & Ash related pump house	Ash Handling Maintenance group
Water Treatment Plant	Offsite Maintenance Group
Cooling Towers including CWPH	Offsite Maintenance Group
Make-up Water pump house	Offsite Maintenance Group
Any other Offsite pumphouse	Offsite Maintenance Group
Switchyard	EMD

Remarks:

- Any left out or new area as per station HOP decision.

6.0 INSPECTION PROCESS AND FREQUENCY

- The inspection is to be carried out by a team comprising of at least two members from the concerned area/group.
- Preliminary/Visual Inspection:

Plant Areas	Internal/External	Committee Team Members	Frequency
All Areas	Internal	Individual Maintenance section executives.	Yearly after COD
All Areas	External	External agency with Individual section executives	Every 5 th year, after 10 years from COD
CHP/Cooling towers	External (Jointly)	External agency with Individual section executives	Every 5 th Year from COD

- The report of preliminary inspection by maintenance group will become additional inputs for the joint preliminary assessment by external expert agency.

- The checklist for inspection is enclosed as Annexure-1. However, the checklist is only indicative and inspection needs to be carried out to cover the total health assessment of structures.
- The frequency of inspection should not be less than as mentioned but can be suitably reduced depending upon the site specific requirements. For difficult access areas, assistance of drones may be taken.
- In addition to the frequency of inspection defined above, any external expert may be involved for additional assessment whenever and wherever required. In case of any specific structure damage or failure, detailed investigation with the required tests may be carried out with any external expert as per requirement.
- Compliance of above inspection system will be reviewed in ORT and ROPR through attached format as Annexure-2

7.0 STATUS OF ASSESSMENT

7.1.1. The status of assessment and repairs should be reviewed at station level in ORTs.

7.1.2. A format will be reviewed in ORT & ROPR (Format as in Annexure2).

8.0 RECTIFICATION

For all significant defects repairs shall be completed as soon as possible. Services of specialists should be taken, if required to ascertain the causes of defects, recommend methods of repair and prevention of recurrence.

A record of such type of repairs to be prepared and should be available as common shared document with MTP Dept.

Some general concepts are given below for rectification of common type of defects.

8.1 Concrete Structures

8.1.1 **Disintegration of concrete:** The extent of disintegration and reasons for the same shall be ascertained first. The surface shall then be cleaned of all loose materials. In case of a large volume of work, recasting are the most appropriate methods. Polymer modified mortar with suitable bond coat before applying PMM is the most suitable for external patch repair upto 50 mm.

8.1.2 **Cracks in Concrete:** Before attempting to repair any cracks, a full investigation should be made to establish the cause of the cracks. Each case should be considered from first principle. any problems should be referred to the designer or to a specialist agency in this field. the type and extent of any repair shall depend upon the nature of crack, type and functional requirements of the member.

Cracks are classified into two types, namely Dormant cracks (which are unlikely to open, close or extent further) and live cracks which may be subjects to further movement. Dormant cracks can be further classified as fine (up to 1.0 mm wide),

wide (above 1.0 mm up to 6 mm) and fracture (more than 6 mm) depending upon the width of the neck. For filling fine cracks (about 0.15 to 1.0 mm wide) cement grouts mixed with polymeric compound are injected. For structurally significant location, injection of epoxy grout shall be resorted. Wide crack on a vertical surface can be repaired by injection method. However, cracks on horizontal surfaces can be repaired by injection or any other simpler method. In case the cracks do not have structural importance, these can be filled, after forming a groove, using flexible sealants, epoxy, or polymeric cement mortar.

If the crack is due to reinforcement corrosion, the simple sealing or such cracks is not recommended as the primary means of repair, as it would not prevent further corrosion and fresh may develop.

Live cracks should be treated as movement joints, these should cater for the anticipated movement, and a suitably dimensioned recess should be cut along the line of crack and then sealed with sealant (flexible type).

For fractured surface, recasting is also undertaken depending upon the extent of damage. In order to minimise the shrinkage in such cases, precautions should be taken to prevent premature drying out and to ensure proper curing.

- 8.1.3 Seepage/Leakage through Concrete:** Location and extent of leakage shall be ascertained before undertaking the repair work. A specialized agency shall be engaged to carry out the work.

The injection of cement grout (mixed with polymeric compound) is the commonly adopted practice. In this method, a nozzle is fixed at the damp/ leakage location and grout is injected under pressure. Detail of the chemical injection method is given in IS 6494. In case the amount of leakage is high and it is not possible to reduce the same, (as in the case of deep basement type structure), the chemical injection method would not be effective. In such cases, the injection of polyurethane foaming compound can be used. This material also solidifies and expands (about 6 times) when it comes in contact with water thus reducing the seepage. However, this material is very costly.

Other methods of rectification can also be followed when a dry working condition can be achieved.

- 8.1.4 Excessive deflection:** Depending upon the effect of deflection (functional, structural and/or aesthetic appearance) of the structure, a scheme shall be prepared in consultation with the designer for rectification of the same.
- 8.1.5 Settlement/differential settlement of the structures:** The rectification scheme shall be prepared after analysing the extent of additional loads and forces on the structures due to this effect and also considering the functional requirements. This may include proper strengthening of the structure. The matter shall be referred to the designer for developing a suitable scheme.
- 8.1.6 Corrosion of reinforcement steel:** After removal of the concrete, the exposed reinforcing bars must be cleaned. The exposed bars should be examined for any excessive reduction in diameter which might impair the strength of the member.

The extent of corrosion can be very variable. For primary tension or compression steel, a 10% reduction in diameter is normally considered to be the upper limit for acceptance. However, small diameter bars, e.g. links, generally show the worst effects of corrosion and the significance of the damage will need to be assessed in each case.

The additional reinforcement should be lapped or anchored on either side of the defective length, for the distance recommended by the relevant code of practice. This also applies to links which are designed to resist shear. It is important that the cover to the reinforcement complies with the code of practice recommendations. Anti-corrosive treatment is to be provided over the reinforcement to avoid any corrosion in the future.

- 8.1.7 **Abrasion of concrete surface:** The surface exposed to regular abrasion shall normally be repaired by guniting. However, for small or localised damage, cement or epoxy mortar can be used. The use of epoxy mortar shall, generally, be restricted due to its prohibited cost.
- 8.1.8 **Expansion joint/gap of isolation joint:** The accumulated dust or any other non-compressible substance shall be cleaned/removed. The damage of damping pads, if any, provided in the isolation joint shall be replaced.

8.2 Steel Structure

- 8.2.1 **Damage to joints due to loosening of permanent bolts or cracks in welds:** In case of cracks in the welded joints, the same shall be rewelded. The loosened nuts shall be tightened, and the missing nuts & bolts shall be provided. If a bolt is corroded, it shall be replaced.
- 8.2.2 **Reduction of cross-sectional area due to rusting (corrosion) or abrasion:** The details of the effective section shall be furnished to the designer to ascertain the safety of the structure and accordingly, a strengthening scheme shall be prepared.
- 8.2.3 **Excessive deflection:** Depending upon the effect of deflection (functional, structural and/or aesthetic appearance) on the structure, a scheme shall be prepared in consultation with the designer for rectification of the same.
- 8.2.4 **Free movement of expansion/sliding joints:** In order to ensure free movement of expansion/sliding joints, the bearings and the slotted holes shall be cleaned of any dust accumulated.
- 8.2.5 **Part cutting of a section/missing member:** The part cut member shall, either be replaced or suitably strengthened. The missing member shall be restored. In case of any problem in accommodating the missing/removed member, the matter shall be referred to the design engineer for suitable alternative arrangement.
- 8.2.6 **Transmission of vibration to the structure from machines or other moving parts.** The source of unbalanced force causing transmission of vibration to the structures shall be ascertained and rectified. However, for strengthening/stiffening of the structure, the matter shall be referred to the design engineer.

- 8.2.7 Maintenance/ Protection of Cable tray: Any coal dust/dust/ debris accumulated on cable tray structure and on its cover must be cleaned periodically. Cable tray should be protected from any ash, coal or dust accumulation with protection sheet of GI or any suitable material. Any structure joint, damaged or missing members of cable tray must be repaired or replaced as per inspection report recommendations time to time. Also, cable tray should be protected in such manner, there should be no waterlogging on its cover or on structural members.
- 8.2.8 Accumulation of debris/ coal /ash particles on structures: In case of debris/ coal/ash accumulations on structures, the same needs to be cleared by making proper scaffolding/approaches. Accumulation of Debris / coal/ash particles needs to be inspected in areas like cable galleries/pipe racks/ conveyor galleries etc.

8.3 Maintenance Painting on metal surfaces

The breakdown of a paint film is a progressive one starting with the topmost coat, gradually going downwards to the primer coat. The object of maintenance paint is to renovate the paint film periodically, checking the progressive breakdown and maintaining the protective and decorative value of the paint film, thus prolonging its effective life. The principles of maintenance painting are given in Table-1. (Reproduced from IS: 1477 Part-II)

Table-1

S No.	Condition of Surface				Cleaning and Painting Procedure
	Rusting	Cracking	Checking	Chalking	
i)	Nil	Nil	Nil	Yes	Clean: rub with sandpaper; apply one reviver coat
ii)	Nil	Nil	Yes/ Fine checking	Nil	Flat down to remove checked films; clean; apply one or two reviver coats
iii)	Nil	Yes. Not Reached the surface below		Nil	Treat as in 8.3.2
iv)	Yes	Nil	Nil	Yes	Treat as in 8.3.2
v)	a. Localized	Nil	Yes	Nil	Treat as in 8.3.2
	b. Localized	Yes	-	-	Treat as in 8.3.2

	c. Localized	-	-	-	Completely remove paint to bare metal and paint with full schedule
vi)	Heavy rusting	-	-	-	Completely remove paint to bare metal and paint with full schedule

The symptoms of paint breakdown are chalking, hairline cracking, deep checking, fine checking, peeling, blistering, and rusting. They may be present either alone or along with other depending upon the degree of breakdown. The breakdown normally starts from the top coat. Generally, when there is chalking, checking and cracking do not appear and the breakdown is a gradual process of erosion. This type of failure is most desirable as the protective value of the film is retained over a longer period and renovation is easier.

The symptoms given above may appear all over the surface or it may appear in certain in patches. Blistering is another symptom of breakdown. Checking, blistering and peeling expose the primer coat and the surface underneath to the weather and results in corrosion. Therefore, such a breakdown needs to be remedied quickly.

If the specification of the painting operation is known, maintenance painting may be easier and more effective as it is possible to renovate by using the same quality of paints as used earlier. If this is not known, maintenance painting has to be carried out on the overall strength of the existing film and renovated by using the best material required to perform under the conditions of exposure.

8.3.2 RENOVATION

The specification of painting in known

If the paint is only chalking and otherwise the film is in good condition, wipe off the chalked film with a wet rag or preferably wash the film with water, remove gloss, allow the water to dry off and apply one coat of finishing all over the surface. if the chalking is in patches, apply one coat at the chalked areas and after it is dry, apply another coat all over the surface.

If fine checking and/or hairline cracking have appeared and the film underneath is firm without symptoms of rust, cut the paint film by wet rubbing, removing all cracked film, wash, allow the water to evaporate and apply one or two coats of (chemical resistant) the finishing paint. If the defects are noticed in certain areas only as patches, apply one coat of the finishing paint on this area and then follow with a coat all over the surface.

If the cracking is deeper, with or without flaking, reaching up to the primer, without any symptoms of rust and the primer coat is in good condition, proceed as follows.

- a) If the defect is all over the surface, cut down the top coats up to the primer by wet rubbing and allow the water to dry off. Touch up the primer coat wherever necessary, and build up the entire film by applying fillers, undercoating and finishing coats as required.
- b) If the defect is noticed only in patches in certain areas, cut down the top coats only in these areas feathering off the top coats, build up a film in these patches ending with one finishing coat and apply a coat of finishing all over the surface.

If blistering with or without flaking is noticed and the primer coat is sound, follow the procedure as (a) and (b).

If rust spots are noticed in certain areas only, with or without any of the other symptoms of breakdown, cut down the film in those spots to the metal, feathering off the topcoats and apply two coats of primer and build up the film following the schedules. Rough sand the entire surface, wash off as necessary and apply a reviver finishing coat.

If (a) the rusting is noticed all over the surface, or (b) rusting is severe or, (c) cracking and blistering has damaged the primer coat exposing the metal and is noticed all over the surface or (d) the paint film has eroded badly; then scrape off the entire paint film to the base metal and repaint.

In case of damage of any metal coating, e.g., galvanised steel or sheet, the same shall be maintained by application of metal spray coating or suitable painting. The type and detail of paint shall be according to the functional requirement, e.g., for structure in contact with water, coal tar epoxy paint would be suitable.

9.0 RECORDS TO BE MAINTAINED

Sl No.	Record Description	Maintained by	Frequency	Remarks
1.	Internal Inspection Report	MTP	1 Year	Reports to be uploaded in webpage of Station MTP
2.	Joint (External) Inspection Report	MTP	5 Years	

10.0 ASSOCIATED DOCUMENTS

Annexure – I: Checklist for Inspection

Annexure – II: Area Wise Inspection Record & Action Plan

Annexure – III: Reference Report

Annexure – I: Checklist for Inspection

CHECKLIST FOR INSPECTION

SN	Condition	Action	Yes/No
a)	Removed/Missing member/Part Cut Members	Visual inspection	
c)	Damage to joints:		
	i) Bolted connection	1. Visual inspection for loosened bolts (if required with spanner)	
		2. Visual inspection of missing lock nuts.	
		3. Visual inspection of missing bolts (uncovered holes if no welding suspected)	
	ii) Welded connection	Inspection/Checking of cracks (if suspected) by dye penetration method.	
d)	Corrosion/rusting/Perforation	Visual inspection	
e)	Abrasion	Visual inspection	
f)	Buckling	Visual inspection (If found , approximate extent of deflection to be recorded) with probable cause of deflection	
g)	painting/coating	Visual inspection to ascertain the:- Condition of primer and finishing coats. e.g. blistering rusting, cracking, checking and chalking.	
h)	Expansion/sliding joints	Visual inspection for checking the dust accumulation in the joints and the free movement of bolts in the slotted holes	

i)	Transmission of vibration (floor adjacent to rotating equipment foundation, crane girder conveyor galleries, etc.	Vibration amplitude shall be measured using a vibration meter.	
		On observation of higher values of vibration amplitude, natural frequency of the structures and the operating frequency of the vibrating source shall be checked to examine the Possibility of resonance, if any.	
k)	Accumulation of Coal dust/Ash on Cable trays	Cleaning of trays in area where accumulation can increase loading of trays during respective unit OH.	
l)	Accumulation of debris/ coal/ash particles	Visual inspection by proper access, especially for cable galleries/pipe racks /conveyor galleries. Help of binocular/telescope/ drone may be taken	
2. Concrete Structure			
a)	Disintegration concrete/ condition.	Visual	
b)	Any abnormal Crack	Location and width of the crack.	
c)	Deflection	Visual inspection	
d)	Seepage (leakage/dampness)	Visual inspection -Type, location and extent of seepage - Ground water table at the time of seepage(When the seepage is due to ground water).	
e)	Settlement/Differential	Checking of reference mark level on settlement the pedestal/column, with the permanent benchmark, with the help of a dumpy level or theodolite.	
f)	Corrosion reinforcement	At apparent location of cracks due to corrosion of reinforcement or at places where rust brown patches are noticed. Covers shall be removed to expose the reinforcement to ascertain the condition of the steel (minimum at 3 locations).	
g)	Abrasion	Visual inspection	

Annexure – II: Area Wise Inspection Record & Action Plan

Area	Structure Type	Last Inspection done	Next Inspection due	Inspection Category (Preliminary /Detailed) Name of Agency conducted inspection	Status of Report	Report Uploaded on Webpage Common Folder (Yes/No)	- Observations - Action Taken - Action to be taken	Target Date
CHP area	Civil							
	Steel							
TG area	Civil							
	Steel							
Boiler area	Civil							
	Steel							
Ash Handling area	Civil							
	Steel							
Water treatment/CT/ MU PH/Other	Civil							
	Steel							
SwitchYard	Civil							
	Steel							

Annexure – III: Reference Report

Categories of Type of Defects observed during one of the Preliminary Investigation:

1. Requirement of Painting almost all over CHP steel structures.



2. TPs near boiler have lot ash deposits on main columns. Like:



3. Concrete reinforcement visibility at some places in TPs and Crusher houses. Like:



4. TP/Crusher column corners have corroded bracings at floors at many locations **because of which the bracing got discontinued and are not connected with the columns**. These bracings needs to be restored and after restoring **300 mm high encasement with concrete** to be done to avoid any future corrosion. Like:



5. **Discontinued/Missing** bracings (1-2 in conv gallery bottom), removed during any material lifting operation:



6. Seepages at roofs of TPs leading to corrosion of beams:



7. Scrap collections at some TPs causing non access and dust accumulations **which will further support the corrosion:**



8. Hydrant leakages causing corrosion:



9. GTU structures need cleaning so that foundation bolts do not fail. Like:



10. GTU platforms and its supports need cleaning regularly and repairs for safety in particular:



11. Various other platforms like Hoist platform, access platforms erected for miscellaneous works ,need painting and foundation strengthening , for safety in particular.



12. Coal dust depositions inside galleries particularly Bottom chord of side trusses at some locations inside galleries/TPs **needs to be cleaned or else corrosion will start or already started.**



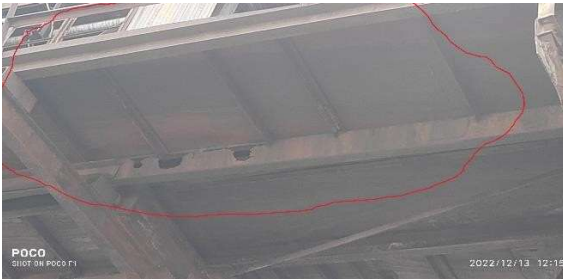
13. Accumulation of coal dust over the pedestals supporting the technological structure needs to be regularly cleaned to avoid corrosion of structure and bolts. At few locations pedestal supporting the structure is damaged which needs to be repaired.



14. Reinforcement of the walls of the underground track hopper is exposed and corroded needs to be repaired or else seepage will start inside the tunnel and corrosion of reinforcement will increase. Exposed reinforcement of beams at column beam junction.



15. Stone picker zone – severe corrosion with pitting was observed in the members at the stone picker zone. Probable cause of corrosion is urination by the workers over the years.

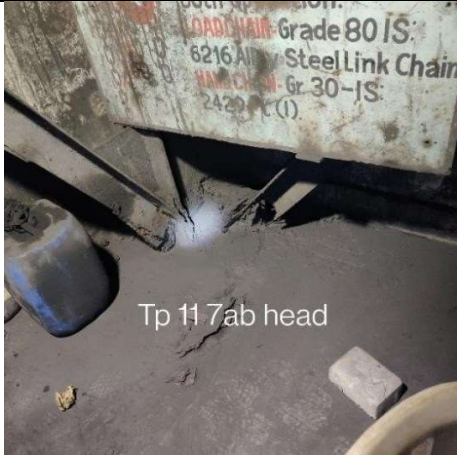


16. Wherever good encasings are done in columns no impact of ageing or water. Like:



Detailed site visit pictures along with description of issues can be recorded as follows:

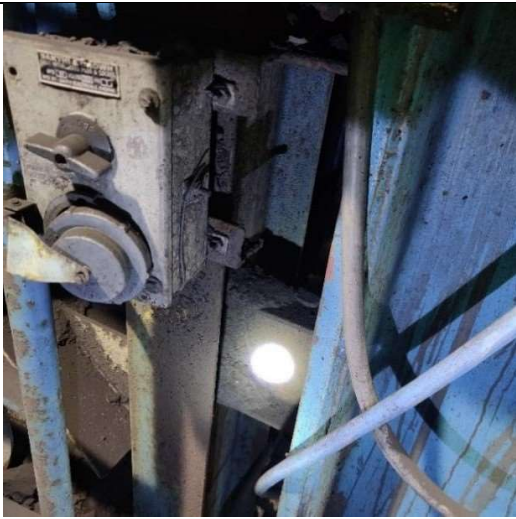
15/12/22: TP11

	
TP11 14A/B tail corner bracing and gusset plate corrosion	TP 11 7AB head corner members corrosion

	
TP 11 concrete encasing vertical cracks	TP 11 roof seepages and beam rusting

Some major noticed Defects could be recorded as:

14/12/22: TP 27 issue of 26AB gallery. This gallery had come down at junction of TP27 in 2018. Same was held in place by new vertical column and cross beam. The Corbel had got broken owing to which this happened. Additional members (I beam sections) were welded between newly erected columns and original columns . Also Gallery structure also welded additionally to TP columns. But now weld cracks noticed in these joints.

	
TP 27 additional corbels /supports develop fresh crack	TP27 additional erected Column with supports

	
TP 27 beam supporting Corbel which had got dislodged.	TP27 ,26AB bottom cross beam which had come down

SOME RECOMMENDATIONS:

A. Crusher House Roof issue:

1. Roof concrete needs to be removed so that sudden failure/accident is averted
2. Strength of roof beams may be assessed by condition assessment by third party if the same members need to be retained and the members which are not found suitable to be replaced with the members envisaged in the original drawing. Considering the situation and to avoid any delay complete roof beam is suggested to be replaced. However instead of roof slab casting, metal deck roof sheeting with purlin and 50 mm thick concreting to be done.

B. 19A/B Sagging:

1. Restoration of conv sagging in line with Kahalgaon repair works carried out on 12A/B in Nov-Dec 22 i.e. connections of bottom chords to be restored with the gusset plates.

C. TP 27 Conv 26A/B issue:

1. Immediate weld strengthening of supports between 26A/B and TP27 structure that were welded, when failure had occurred.
2. Revival of Corbel supports of 26AB and then the conveyor to be rested on the corbel as envisaged in the erection drawing as the supporting columns are not provided with any foundation hence not safe.

Generic to all of CHP:

1. Nomenclature (Painted) at TP/Buildings/conveyor/TRs columns needed for specific identification of defects. This will also help in quantification of work.
2. Record of Structure repairs in case of failures must be maintained so that History is available, and condition can be assessed at later stages.

3. Station is checking structures through its CFT. Its activity should be completed in designated schedule and action plan made.
4. For conveyors/TPs higher than 20 m use of Drone may be explored. Once a year contract may be initiated for such structure checking with optical camera.
5. Stone picker zones to be monitored specially bottom chord members which gets corroded probably due to urination of the workers. In case corrosion is observed, immediate action to be taken like replacement of members or
6. The beams/bracings which have been curtailed due to site requirement and the members which have been discontinued/disconnected due to extensive corrosion must be restored.
7. The connection either welded or bolted which have got disconnected must be restored (Like for column bracings at floors) .
8. Regular clearing of accumulation of water/coal/ash over the members to avoid any future corrosion.
9. After restoration of the column & bracings connection in the floors of crusher house/TP's at the floor location columns needs to be encased upto a height of 300 mm to avoid any future corrosion, with slope.
10. Complete painting as per OGN to be done for all the steel structures.
11. Exposed reinforcement of the RCC structures needs to be applied with alkaline based rust remover for removing of rust, anti-corrosive treatment over the r/f, application of bond coat for adhesive between old concrete and new concrete and then apply PMM for a max thickness of 50 mm. In case cracks are observed than polymer grouting to be done before application of bond coat for PMM.
12. Housekeeping needs to be done on regularly basis. No coal deposit in the conveyor galleries, particularly bottom chord of trusses.
13. Hydrant drain routing is very important. At most TPs severe corrosion at stairs and columns here these hydrants are there. These hydrant drains need to be away from structures.
14. Damaged cladding sheets and window glazing's should be replaced.
15. Arresting roof seepages in buildings. Elastomeric roof water proofing to be done (as per present NTPC practise) over the RCC roofs to avoid any seepages in the RCC roof slabs.
16. Scrap removal from sites so that all structures are visible.
17. Conveyor Take up structures and other working platform structures erected in buildings:
 - a. Cleaning of platforms and repairs.
 - b. Clearing of coal deposits at foundations to check foundation bolts and repair if required.

REVISION INDEX

(for changes/revision)

Sl. No.	Rev. no.	Date	Details	Approved by
1	1	19.04.2006		GM(Simhadri)
2	2	19.04.2008		GM(Simhadri)
3	3	14.06.2010		GM(Simhadri)
4	4	March 2013		BUH(Simhadri)
5	5	Jan 2020		CGM (Simhadri)
6	6	Jan 2021		BUH(Simhadri)
7	7	Feb 2023	Structure list updated	BUH(Simhadri)
8	8	June 2023	Revised as per Latest OGN	BUH(Simhadri)

DISTRIBUTION LIST

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